# KUNAL TYAGI

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## SUMMARY

Learning and developing skills in the domain of Data Science, Machine Learning, and Full-Stack Web Development. Strong analytical and problem-solving abilities combined with excellent English communication skills, with a proven ability to manage responsibilities and ensure smooth execution of tasks.

## EDUCATION

**Manipal University Jaipur** — *Jaipur, Rajasthan*
*Bachelor of Technology* — *Oct 2024 – May 2028*

- Major in Computer Science and Engineering (Core)
- CGPA: 7.5 (as of 4th Semester)
- Relevant Coursework: OOP, DSA, Operating Systems, Computer Architecture, Computer Networks, DBMS, High Performance Computing

## LEADERSHIP & EXPERIENCE

**Elite Coders** — *Remote*
*Open Source Contributor* — *June 2026 – August 2026*

- Contributed to open-source repositories by fixing bugs, improving documentation, and implementing new features under the guidance of maintainers.
- Wrote clean, well-documented code and participated in peer code reviews to improve project quality and maintainability.

**Future Interns** — *Jaipur, Rajasthan*
*Full Stack Web Development Intern (Remote)* — *March 2026 – April 2026*

- Worked on full-stack web development projects involving frontend and backend integration.
- Applied concepts of APIs, databases, and responsive UI design in real-world tasks.
- Collaborated in a remote team environment and followed structured development practices.

## PROJECTS

**ML-Powered Credit Card Fraud Detection System | Python, Scikit-learn, XGBoost, Pandas**

- Developed an end-to-end machine learning system to detect potentially fraudulent credit card transactions from highly imbalanced transaction data.
- Performed data preprocessing, exploratory data analysis, feature engineering, and class-imbalance handling using techniques such as SMOTE and class weighting.
- Trained and compared Logistic Regression, Random Forest, and XGBoost classification models to identify fraudulent transactions.
- Evaluated models using Precision, Recall, F1-score, ROC-AUC, and PR-AUC, prioritizing fraud detection performance over raw accuracy.
- Built a prediction pipeline that assigns fraud probabilities to new transactions and flags high-risk transactions for further review.
- Visualized transaction patterns and model performance using confusion matrices, feature-importance plots, and evaluation dashboards.

**Automated Precision Supply Chain | Python, Machine Learning, React, SQL**

- Developed an intelligent supply-chain management platform to analyze inventory, sales, and supplier data and support data-driven procurement decisions.
- Implemented data preprocessing, exploratory data analysis, and feature engineering on historical sales and inventory data to identify demand trends and supply-chain risks.
- Developed demand forecasting models to predict future product requirements and identify potential stockout and overstock conditions.
- Implemented inventory optimization logic using predicted demand, current stock levels, safety stock, and supplier lead times to generate automated reorder recommendations.
- Built an interactive React dashboard to visualize inventory levels, demand trends, supplier performance, alerts, and replenishment recommendations.

**Sales Analytics & Revenue Forecasting | Python, SQL, Pandas, Power BI**

- Developed an end-to-end sales analytics solution to analyze revenue, product performance, customer segments, regional sales, and purchasing trends using historical transaction data.
- Performed data cleaning, exploratory data analysis, KPI development, and SQL-based analysis to identify revenue drivers, underperforming products, and high-value customer segments.
- Built interactive dashboards tracking revenue, profit, sales growth, average order value, product performance, and regional performance to support data-driven business decisions.
- Implemented time-series forecasting to estimate future sales and identify seasonal demand patterns and revenue trends.
- Generated actionable insights and recommendations for inventory planning, product prioritization, and sales strategy.

**Mini CRM – Customer Relationship Management System | React, Node.js, Express, MongoDB**

- Developed a full-stack CRM platform to manage customer profiles, leads, sales opportunities, interactions, and follow-up activities through a centralized interface.
- Implemented CRUD operations, lead management, customer search/filtering, task tracking, and sales-pipeline management.
- Designed RESTful APIs and integrated a MongoDB database for persistent storage and efficient retrieval of customer and sales data.
- Built an interactive dashboard providing visibility into lead status, conversion progress, customer activity, and sales performance.

## CERTIFICATIONS

**AI in Healthcare – IEEE Research Paper**

- Authoring an IEEE-format research paper on Synthetic Clinical Data Generation Using AI: A Privacy-Preserving Approach for Medical Research.
- Explored regression-based and representation learning approaches for clinical outcome prediction and risk assessment.
- Applied machine learning techniques to healthcare data analysis and prediction.

## SKILLS

**Programming Languages:** C, C++, Python, SQL, MongoDB
**Web Development:** HTML, CSS, JavaScript, React, Node.js, Next.js, Tailwind CSS
**Data Analysis & ML:** EDA, Regression Analysis, NumPy, Pandas, Scikit-learn, Logistic Regression, Matplotlib, Seaborn
**Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook
**Languages:** English (Fluent), Hindi